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IEGP

INDUCTION MOTORS

DOC. NO. : IEGP-6351-8531-SP-000-1001

PROJECT NAME: PMC SERVICES FOR ETHYLENE GLYCOL PROJECT AT PARADIP REFINERY

TOYO JOB NO. : 6351

OWNER: INDIAN OIL CORPORATION LIMITED.

PMC : TOYO ENGINEERING INDIA PRIVATE LIMITED.

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TOYO ENGINEERING INDIA PRIVATE LIMITED
MUMBAI, INDIA

TOYO ENGINEERING INDIA PVT. LTD.	**INDUCTION MOTORS**	DOC NO. : IEGP-6351-8531-SP-000-1001
ISSUED : 09.02.2018	**IEGP**	PAGE 1 OF 4

REVISION HISTORY	
REV.	**DESCRIPTION AND/OR REVISED SHEETS**
I1	Revised as per IOCL Comments
I2	Issued for Tender

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1.0 General

2.0 Site Conditions

3.0 Utilization Voltages

4.0 Specific Requirements for HV & VFD operated LV motors

5.0 Additional requirements for VFD driven Motors

1.0 General

General requirements for the design, manufacture and testing for Induction Motors are specified in this specification. PDRP specification (Document no. PDRP-8410-SP-007 Rev-F7) also forms a part of this specification.

2.0 Site Conditions

The equipment shall be suitable for following site condition.

- Min. Temperature (T min) - 11.3°C
- Max. Temperature (T max) - 42.4°C
- Design Temperature - 45°C
- Relative humidity at T max. - 99.7%
- Seismic zone – Zone 3

3.0 Utilization Voltages

Unless otherwise specified, utilization voltages of the motors shall be as below,

Motors rated >4 MW	:	11kV ±6%, 3-Phase, 50Hz ±3%, AC Supply
Motors rated >160≤4MW	:	6.6kV ±6%, 3-Phase, 50Hz ±3%, AC Supply
Motors rated 0.18kW-160kW	:	415V ±6%, 3-Phase, 50Hz ±3%, AC Supply
Motors rated <0.18kW	:	240V ±6%, 1-Phase, 50Hz ±3%, AC Supply

4.0 Specific Requirements for HV & VFD operated LV motors

Induced voltage at the shaft end of the motor at no load shall not exceed 250mV r.m.s. for roller and ball bearings and 400 mV r.m.s for sleeve bearings. The non-driving end bearing shall be insulated from the motor frame to avoid circulating current. The insulated bearing end shield or pedestal shall bear a prominent warning and manufacturer shall provide detailed drawing showing insulation arrangement.

5.0 Additional requirements for VFD driven Motors

(1) The motor shall be suitable for the current waveforms produced by the power supply including the harmonics generated by the drive.

(2) The motor shall be designed to operate continuously at any speed over the range 1-100% of rated speed.

(3) The permitted voltage variation should take into account the steady state voltage drop across the AC drive.

(4) Motors required to be transferred to DOL by-pass mode shall be rated for specified variations in line voltage, frequency and starting current limitations.

(5) The motor shall be constructed to withstand torque pulsations resulting from harmonics generated by the solid state power supply.

(6) The motor insulation shall be designed to accept the applied voltage wave form, within the Vpeak and dv/dt limits as per IEC-61800-4.

IndianOil	PARADIP REFINERY PROJECT PROJECT SPECIFICATION	PDRP-8410-SP-0007 PAGE : 1 of 17 REV : F3
	INDUCTION MOTORS	

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Changes in this document are highlighted.

Section	Change Description
2	IS 325 referenced to align with ATI-ALL-00041
4.1	4.0MW limitation, to align with ATI-ALL-00041
6.10	Amended to align with ATI-ALL-00007
6.11	Amended to align with ATI-ALL-00034

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F7 I 09.02.2018 I Issued For Tender I ANP

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ATTACHMENTS

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ATTACHMENT 2 DATA REQUIRED AFTER PLACEMENT OF ORDER

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ATTACHMENT 3 LV INDUCTION MOTOR DATA SHEET

ATTACHMENT 4 HV INDUCTION MOTOR DATA SHEET

1.0 FOREWORD

The Vendor shall ensure that all equipment in his scope of supply complies with all applicable statutory requirements, ordinances, standards, codes, local rules and regulations.

2.0 APPLICABLE SPECIFICATIONS

The motors shall comply with the following specification /standards.

IEC 60034	Electrical Rotating Machines
API 670	Vibration, Axial Position, and Bearing Temperature Monitoring Systems
~~API 678~~	~~Accelerometer-Based Vibration Monitoring System~~ F7
IS 325	Three Phase Induction Motors

Project Spec. No.	Title
PDRP-8410-SP-0011	Guards for Machinery
PDRP-8550-SP-0006	Instrumentation for Packaged Units
PDRP-8440-SP-0006	Protective Painting System
PDRP-8440-SP-0013	Equipment Noise Control Specification
PDRP-8531-SP-0001	Electrical Standard
PDRP-8820-SP-0001	Basic Engineering Design data

The Vendor shall ensure that all equipment in his scope of supply complies with all applicable statutory requirements, standards, codes, local rules and regulations.

3.0 GENERAL

3.1 Compliance with this Specification does not relieve the Vendor of the responsibility for furnishing equipment of proper design and construction and fully suitable for all specified operating conditions.

3.2 Equipment offered shall satisfy the following minimum service and manufacturing experience requirements:

a. It shall be identical or validly similar in power rating, speed, model, frame, voltage, mechanical and electrical design and materials as compared with at least two units produced at the proposed manufacturing plant One of which must have had at least one year's satisfactory and continuous operation without any interruption at the time of enquiry issue.

b. Equipment manufactured under licence, in addition to satisfying the requirements above shall also have the licensor's written guarantee.

3.3 The equipment (including auxiliaries) covered by this Specification shall be designed and constructed for a minimum service life of 20 years and at least 3 years of uninterrupted operation. Vendor shall recognize that this is a design criteria.

4.0 SERVICE CONDITIONS

4.1 General

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, Standard Efficiency,

Unless stated otherwise all motors shall be squirrel cage induction type, single speed and started direct on line.

Motor cooling will fall into the following categories:

TEFC	Totally enclosed fan cooled
CACA	Cooling air to cooling air exchanger
CACW	Cooling air to cooling water exchanger

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~~(as indicated on the motor data sheet)~~

Motors up to 160 kW shall be powered at 415 volts 3phase 50 Hz.

~~(Refer Note 1 below~~

Motors over 160 kW up to 4.0MW shall be powered at 6600 volts 3 phase 50 Hz.

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Motors over 4.0MW shall be powered at 11000 volts 3 phase 50Hz.

Motors shall be designed for voltage variations of plus or minus 6% and frequency variation of plus or minus 3%

In all cases the motor shall be sized by the driven equipment Vendor.

In all cases the motor is required to start and run up the driven equipment at 80% of rated voltage, and still maintain at least a 10% margin over the driven equipment torque at all points on the speed torque curve.

The equipment Vendor, at time of quotation, is required to supply a speed / torque curve showing:

- The driven equipment torque requirement
- The motor torque at 80% of rated voltage
- The motor torque at 100% of rated voltage

4.2 On Site Conditions

The on site service conditions are coastal, tropical and corrosive as defined in project Specification PDRP-8820-SP-0001 Basic Engineering Design Data. The equipment (including Auxiliaries) shall be suitable for unsheltered outdoor operation, unless indicated otherwise.

~~1) VFD driven motors for gear pump in Extruder package shall be suitable for voltage of 3300V +- 6%~~

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All Electric motors shall be rated for the project design ambient of ~~43ºC~~. 45 Deg.C F7

4.3 Motor Enclosure and Area Classification

For each specified service, the area classification and motor enclosure applicable shall be as specified on the motor data sheet and be in accordance with the applicable standards.

The enclosure integrity of all motors and their cable termination boxes shall be IP 55 minimum.

Motors shall be certified suitable for the area in which they are installed as detailed in Project Specification PDRP-8531-SP-0001 Electrical Standard Section 7.2.

5.0 WINDINGS AND INSULATION

5.1 All motor winding insulation shall be class F vacuum impregnated, but with temperature rise restricted to class B

5.2 All six ends of the windings shall be brought out to the main terminal box.

5.3 The windings of all low voltage motors below 30kW shall be guaranteed for use, and for standing idle in the environmental conditions indicated without the use of anti condensation heaters.

5.4 All low voltage induction motors 30kW and above and all medium voltage motors shall be fitted with 240 volt AC space heaters wired to a separate junction box.

5.5 The windings of all medium voltage motors shall be suitable for switching using vacuum circuit breakers. Vendor to confirm compatibility.

5.6 Thermistors shall be provided in all VFD operated LV Motors.

6.0 GENERAL CHARACTERISTICS

6.1 Motors shall withstand all transient torques associated with any transient interruption in supply voltage, without detriment.

6.2 When reacceleration is specified, the motor shall be capable of recovering normal operation with voltage restoration to 100 per cent after total disconnection. The percentage value of residual voltage against which the motor may be reconnected to the supply and any minimum allowable associated period of supply interruption shall be advised to IOCL.

6.3 The safe stalled time for cold and hot conditions shall be stated. In windings with underrated insulation e.g. Class 'F' with a Class 'B' permitted rise, the higher temperature is allowable.

6.4 For Ex e motors, the te time shall be given in place of the stalled time, but for Ex N machines the Allowable Run-up Time (ART) shall be given in addition to the safe stalled time.

6.5 The ART is that time for a motor to complete one start when coupled to a load with the run-up characteristic of the actual load, but where the total inertia is such that when full speed is reached, a part of the motor has reached the limiting 'T' classification temperature.

6.6 The ART and safe stalling times shall be given at the specified terminal voltage and assumes that the initial motor temperatures shall be the working temperatures at rated load.

6.7 The breakdown torque for all motors shall not be less than 200% of full load torque unless specified otherwise.

6.8 The maximum full load slip as percentage of the synchronous speed at full rated voltage and frequency shall not exceed the following:

- Motors smaller than 15 kW - 5%
- Motors 15 kW and larger - 3%

6.9 It shall be assumed that the motors will be circuit breaker connected to switchgear having the following fault levels:-

- At 415 volts 50 kA for 1 second
- At 6600 and 11000 volts 31.5 kA for 1 second

The equipment Vendor shall confirm compatibility.

6.10 The starting locked rotor current of all motors shall be restricted as follows, with no upwards tolerance:

- Low voltage up to 0.415kV machines to 6.5 times full load current
- Medium voltage up to 11kV machines rated upto and including 3.5MW restriction to 4.5 times full load current is preferred. Where the manufacturer is unable to achieve this minimum a starting current of 5.5 times full load current may be considered subject to ~~EPCm-7~~ system analysis and approval.
- Medium voltage up to 11kV machines, rated above 3.5MW and up to and including 4MW, restriction to 4.5 times full load current.
- For 11kV motors rated above 4.0MW fed via captive transformer, in each case the type of starter shall be subject to ~~EPCm-7~~ system analysis and approval.
- IOCLs approval is required for any motor for which VSD or soft starter is considered only for the purpose of limiting the starting current to less than 4.5 times full load current.

6.11 All 6.6kV and 11kV motors rated above 750kW or with VSD shall be fitted with:

For 415V motors winding, bearing RTDs, bearing vibration monitor are not required

- Winding RTDs, three per phase
- Bearing RTDs, two per sleeve bearing

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- Bearing vibration monitors ~~as indicated on the motor data sheet~~. F7

6.12 Motors of 1500 kW and above shall be fitted with current transformers to provide differential protection.

6.13 Motor heaters and the above protective devices shall be wired down to a series of single function auxiliary junction boxes, for export connection purposes.

6.14 Unless stated otherwise, the main cable entry box on all horizontal motors shall be located on the ~~right~~ left hand side of the machine when viewed from the non drive end.

6.15 All main cable boxes shall be of oversize proportions. The main cable box shall be able to be turned through 90 degrees, such as to permit entry from below if ever necessary. Unless stated otherwise, the main feeder cable shall enter the machine in a horizontal direction from the non drive end.

6.16 All medium voltage motors for hazardous area shall be fitted with half inch NPT connections to facilitate pre-start purging ~~by IOCL~~.

Refer point 6.17 mentioned below

7.0 MATERIALS

7.1 Motor frames shall be constructed of ~~cast iron~~ or carbon steel unless otherwise specified. They shall be suitable for withstanding all forces imposed upon them during starting, normal operation and sudden stops.

7.2 Fan covers shall be made of cast iron, cast steel, steel plate or sheet steel. All motor fans shall be made of a non sparking material.

7.3 Sheet metal covers used to form air passages over the motor enclosure shall be made of carbon steel or stainless steel.

7.4 All bolts, studs and other fastening devices on the outside of the motor enclosure shall be made of corrosion-resistant material or be plated or treated with corrosion-resistant material.

7.5 Air-to-air heat exchanger tubes for tube-type motors shall be made of copper, copper-base alloy, aluminium alloy (with maximum of 0.2% copper) or stainless steel. Copper and copper base alloys shall not be used in an ammonia polluted atmosphere. See section ~~14.0~~ sec 15.0 for TEWAC Motors. The air to water heat exchanger on water cooled motors shall be of cupro-nickel material.

7.6 Asbestos shall not be used in any form.

7.7 All medium voltage motors shall have forged steel shafts.

8.0 WINDING AND CABLE TERMINATIONS

8.1 A M10 brass earthing stud with nut, a cadmium plated carbon steel spring washer and a brass plain washer shall be provided.

6.17 Main cable boxes/power terminal boxes of MV motors shall preferably be phase segregated type, suitable to withstand the specified fault level for 0.2 sec.

heat shrinkable kits only except for imported motors where

8.2 Feeder cables to all 415 volt motors shall, in all cases be terminated by way of crimp type termination lugs, which are required to be fitted in place by the equipment Vendor, to suit ~~IOCL's~~ cables. are acceptable

8.3 Feeder cables to all medium voltage motors shall in all cases be terminated by means of plug in elbow type stress relieving connectors (

Elastimould or Pfisterer or equal). The equipment Vendor is required to supply both the male and female parts of the termination to suit ~~IOCL~~ cables.

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8.4 All cable entries shall be bored and threaded ISO metric 1.5 mm constant pitch by the Vendor.

shall be supplied by ~~LSTK1/LSTK2 contractor~~ Purchaser.

8.5 All cable termination glands ~~will, in all cases be supplied by IOCL.~~

8.6 ~~IOCL will advise cable details and cable entry details by means of the motor data sheet.~~

9.0 BEARINGS AND LUBRICATION

9.1 Ball and Roller Bearings

9.1.1 Unless indicated otherwise, the use of ball and roller bearings shall be limited to motors below 1000 kW. All L.V. motors shall have greased for life ball or roller bearings. Medium voltage motors shall be provided with the bearing type as indicated on the motor data sheet.

9.1.2 Unless stated otherwise, for motors rated 1000 kW and above, bearings shall be oil lubricated sleeve bearings.

9.1.3 Horizontal motors with axially unrestrained roller bearings shall be provided with a thrust shoulder, to permit uncoupled testing or running in.

9.1.4 Motors specified for belt or chain drive shall have radial load bearing capacity adequate for the service. The quotation shall state any drive limitations.

9.2 Sleeve Bearings

9.2.1 Pressure lubricated bearings shall be supplied with oil from the driven machine lubrication system, unless specified otherwise.

9.2.2 Horizontal sleeve bearing motors shall preferably have adequate thrust bearing capacity to eliminate the need for limited-end-float couplings.

9.2.3 All horizontal motors with sleeve bearings shall be provided with a thrust location, to permit uncoupled testing or running in.

9.2.4 An indicator shall be provided, to show allowable limits of rotor movement.

9.3 Lubricants

For pressure-fed oil systems, where oil is supplied from the driven machine lubrication system, the motor Vendor shall ensure that the driven machine lube oil is suitable for the motor bearings and shall state the flow rate, temperature and pressure requirements.

9.4 Limited End Float

When limited-end float couplings are required, rotors shall have a total end-float of 1/2" (12 mm) minimum. The running position of the rotor shall be within 3/32" (2.4 mm) of the stator geometric centre.

The end-float limitation shall be stated in the quotation and on the motor drawings.

9.5 End Thrust

The thrust bearings shall be suitable for the maximum axial thrust (up and down) imposed by the driven equipment.

9.6 Bearing Insulation

The requirements for insulation shall include any bearing oil supply piping.

9.7 Lube Oil Piping

9.7.1 All piping with its mounted appurtenances within the confines of the base area shall be furnished terminating with flanged connections at the edge of the base.

9.7.2 All piping shall be type 304 seamless stainless steel with stainless steel weld neck flanges. All piping welds shall have full penetration. An oil flow indicator shall be provided in each bearing oil drain line. The indicator shall be a non restrictive-flow bull's-eye type and made of steel.

10.0 ROTOR

10.1 Shaft extensions shall have a machined keyway, complete with key (taped to the shaft for shipment) to receive the coupling half. For large motors, shaft extensions with taper fit are preferred.

10.2 Flexible couplings for use with horizontal sleeve bearing motors without thrust or location bearings, shall be of the limited-end-float type. The end-float shall not exceed 1/4" (6 mm).

10.3 Motor half couplings, unless specified otherwise, shall be supplied rough bored by the driven equipment Vendor, for final machining and fitting by the motor Vendor.

10.4 Unless specified otherwise, coupling guards shall be supplied by the driven equipment Vendor and shall meet the requirements of Project Specification No. PDRP-8410-SP-0011 ‘’Guards for Machinery.’’

10.5 Cooling fan fundamental blade frequencies and six harmonics shall be 10% away from the motor operating speed and any other sources of excitation.

10.6 The rotor shall be of squirrel cage type, dynamically balanced to provide a low vibration level and long service life for the bearings. Die cast aluminum rotors for motors in hazardous areas may be accepted provided the same are type tested and approved by competent authorities. F7

11.0 NOISE

Noise levels shall be in accordance with the requirements of Project Specification No.PDRP-8440-SP-0013 ''Equipment Noise Control Specification

The extra cost of silencers or other measures proposed, if required to meet the noise limitations, shall be highlighted in the quotation.

12.0 VIBRATION

12.1 The maximum permissible unfiltered vibration limits in a radial direction as measured on the bearing housing shall be 2 mm/sec Peak when bolted to a rigid mounting or 1.25 mm/sec Peak when on an elastic support (rubber, spring, etc).

12.2 Two pole cage-induction motors above 100 kW shall be subject to a broad band vibration analysis. The contribution of twice the supply frequency 100 Hz or 120 Hz to the overall vibration shall not exceed 1 mm/sec Peak.

12.3 When a non-contacting shaft vibration monitoring system is specified, it shall be in accordance with API Standard 670 and the Material Requisition. During the shop test, the double amplitude of vibration, in any plane measured on the shaft adjacent and relative to each radial bearing, shall not exceed the following:

1500 RPM and above	0.002" (0.051 mm)
below 1500 RPM	0.0025" (0.064 mm)

If the existence of electrical run-out is demonstrated, the run-out value up to a maximum of 0.00025" (0.006 mm) may be added to the permissible double amplitude.

12.4 ~~When an accelerometer-based vibration monitoring system is specified it shall be in accordance with API 678.~~ F7

13.0 TESTS

13.1 Each motor shall be given shop tests to demonstrate its freedom from mechanical or electrical defects and prove a satisfactory performance, in accordance with the applicable standards. The type of test, i.e. "complete" or "abbreviated", together with any additional test requirements, shall be specified on the motor data sheet.

13.2 Two pole cage induction motors above 100 kW shall be bolted to a rigid bed plate or foundation for vibration measurements during the shop running test.

13.3 Soft feet check: Before the mechanical run test, motor “soft feet” shall be checked. Micrometer reading shall not exceed 0.02 mm during this

verification. This check shall be carried out at each motor foot, with the other feet in tight condition. "Soft feet" checking shall be performed after the motor alignment and shimming were completed and motor was fixed to the test base securely.

14.0 ANCILLARY EQUIPMENT

Stainless steel nameplates shall be attached to the motor casing with stainless steel pins. IOCL's Equipment Item No. shall be shown on a separate plate.

15.0 TOTALLY ENCLOSED WATER-AIR-COOLED MOTORS

15.1 For totally enclosed water-air-cooled motors, pedestal-mounted coolers or coolers mounted underneath the windings are preferred. For this type of cooler location, the cooler may be of single-tube construction.

15.2 Coolers mounted above the motor windings shall be of double-tube construction. The annular space between the tube walls shall be drained to an inspection area outside the housing.

15.3 Cooler tubes shall be of corrosion resistant metal, suitable for the cooling water used. The interior of the motor shall be baffled or otherwise constructed to prevent cooler-tube leakage from directly striking the motor windings. The cooler shall be provided with a low-point drain.

15.4 The interior of the motor base shall be so constructed that cooler leakage will collect and drain from the motor before reaching the level of the windings. A liquid seal shall be provided for drain holes in pressurised motors.

15.5 A cooling water flow-sensing device for mounting in the water supply piping to each motor shall be provided. The device shall have a local calibrated indicator plus switches activated by low and high flows. The high flow shall actuate an alarm. The low switch shall be supplied with the motor manufacturer's recommended setting below which the motor would overheat and winding damage would result within 15 minutes of low flow occurrence.

16.0 VERTICAL MOTORS

16.1 The bearings shall be grease lubricated unless specified otherwise.

16.2 The shaft shall be solid type unless specified otherwise.

16.3 A weather protecting cover shall be provided on the non-driving end to prevent direct entrance of rain, snow or sleet.

16.4 A non-reverse ratchet to prevent reverse rotation shall be provided for all applications where the reversal of rotation would cause damage to the driven equipment. For flameproof (Exd), increased safety (Exe) and type N

protection (ExN) motors the ratchet device shall be of a non-sparking material.

17.0 SPACE HEATERS

Space heaters shall be provided for LV and MV motors as identified in para 5.4. The location and capacity of the heaters shall be adequate to prevent condensation when the motor is not in operation. Heater leads shall be brought out to a separate terminal box furnished by the Vendor. Space heater surface temperatures shall not exceed the gas mixture temperature class 'maximum surface temperature' specified on the Material Requisition, but shall not exceed 200ºC in an ambient temperature of ~~43 °C~~ 45 Deg.C

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18.0 VENTILATION GUARDS

Open motors shall have corrosion resistant guard screens on all ventilation openings.

19.0 INSTRUMENTATION AND PROTECTIVE DEVICES

19.1 Controls and Instrumentation and its installation shall conform to project Specification PDRP-8550-SP-0006 Instrumentation for Packaged Units.

19.2 As a minimum, motors rated 1000 kW and above and motors of lower rated power but for use with special purpose machinery (e.g. compressors), shall be provided with:

a. Bearing high temperature alarm (each bearing).

b. Bearing temperature indicator (each bearing).

c Bearing housings shall have a provision for mounting removable non-contacting vibration probes in accordance with API 670 Standard. The probes and proximitors shall also be

d. Stator winding high temperature alarm.

e Stator winding high temperature shut-down. (The shut-down temperature shall not exceed the insulation temperature limit).

19.3 Motors with water-air or air to air cooling shall have instrumentation and protection devices to indicate the condition of the equipment and provide the necessary protection instrumentation to alarm and shutdown where necessary. Details to be given in the motor data sheet.

19.4 Differential Protection

19.5 If differential protection is required, matched current transformers shall be provided as specified in the motor data sheet. The neutral terminal box shall

6 CTs to be supplied , 3 CTs at neutral box and 3 CTs at main terminal box and wiring is to be done till motor terminal box.

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and line side F7

accommodate three ring type current transformers. All neutral primary circuits through the transformers shall be completed by the motor Vendor.

19.6 Current transformer secondary leads shall be connected to a terminal strip mounted within a separate external terminal box. The terminal strip shall have links arranged to permit short circuiting the current transformer secondary windings.

20.0 TROPICALISATION

When tropicalisation treatment is specified, all internal parts which are subject to attack by moisture, fungus or insects, shall be treated with polyurethane or equivalent coating and fungicides to inhibit such attacks. Corrosion resistant materials and moisture resistant insulation shall be provided.

21.0 SPARE PARTS

Quotations shall state separate break-down price for spare parts required for commissioning and for two years operation.

22.0 PAINTING

The paint for all exterior surfaces shall be suitable for the environment specified. The paint, preparation and painting procedures shall be submitted for review and approved by IOCL.

Finish colour to be as Project Specification PDRP-8440-SP-0006 Protective Painting System.

23.0 INSPECTION

Replace IOCL with IOCL/PMC/LSTK F7

23.1 Whenever the specifications, Material Requisition or Purchase Order calls for certain shop inspections and test to be witnessed, observed, or performed by a IOCL representative, the motor Vendor shall provide sufficient advance notice to IOCL prior to each inspection or test as per purchase order requirements. At all other times IOCL's representative shall, upon providing similar advance notice have access to all Vendor and Sub-Vendor plants where work on or testing of the equipment is in progress.

23.2 The motor Vendor shall be responsible for notifying all Sub-Vendors of IOCL's inspection requirements.

23.3 IOCL will specify the extent of his participation in the testing programme.

23.4 'Witness' means that a hold shall be applied to the production schedule and the test carried out with IOCL in attendance. It usually implies a double test. Because of the hold and the repetitive test, a charge is made for a witness test.

23.5 'Observed' means that IOCL requires notification of the test timing. However, the test is performed as scheduled, and if IOCL is not present, the Vendor can proceed to the next step. Since only one test is scheduled, IOCL should expect to be in the factory longer than for a witness test. Normally, no additional charge is made for an observed test.

23.6 The Vendor shall provide a quality control plan showing the proposed programme of inspection and testing and which items are to be witnessed or observed, in his bid. This plan will be reviewed and approved by IOCL. IOCL may advise any required amendments. When requested, the Vendor shall provide a copy of his quality control procedures for IOCL's review.

23.7 IOCL's inspector shall be informed of any defects noticed during manufacture. The Vendor shall also inform IOCL and obtain his permission before proceeding with any repairs which may affect equipment operation, integrity or interchangeability. Repair procedures shall be approved by IOCL before rectification. Rejections made by IOCL shall be final.

23.8 Unless otherwise specified, the first motor of an identical series shall be subjected to a full witnessed works test in accordance with the applicable international design standard and each subsequent machine of the series to an abbreviated test and provided with a test certificate.

23.9 Acceptance by the inspector shall in no way release the motor Vendor from the guarantees as to the material, workmanship and performance of the equipment.

24.0 PREPARATION FOR SHIPMENT

The preparation for shipment and the type and mode of shipment shall be in accordance with the purchase order.

25.0 GUARANTEE AND WARRANTY

The guarantee and warranty shall be in accordance with the purchase order.

26.0 VENDOR'S DATA

26.1 Vendor data requirements are as defined in the inquiry and material requisition documents. T he following attachment 1 and 2 provide additional specific requirements to be complied by Vendor.

26.2 All data and information provided during the proposal stage shall also be valid for the contract and "as built " stages of the contract.

26.3 Number, procedure and quantity of drawings and documents required shall be as specified in material requisition, together with a mutually agreed schedule for obtaining submission and approval.

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26.4 Completed "as built" ~~purchaser's~~ data sheets and performance curves shall be provided by the Vendor.

26.5 The technical data manual shall be provided. Format and content shall comply with the requirements of the purchase order.

ATTACHMENT 1
INFORMATION TO BE SUPPLIED WITH THE QUOTATION BY THE VENDOR

Information required with the proposal shall include all information listed in inquiry and material requisition together with the following additional data:

1. Re-acceleration time, when re-acceleration is specified.
2. L10 rated life of anti-friction bearings.
3. For all motors.

   Allowable time at locked rotor conditions with rated voltage for:

   - Cold start: the motor at rated ambient temperature.
   - Hot start: the motor at allowable full-load temperature rise at rated ambient temperature.

   Required cool-down time after third start.

   Maximum number of starts per day recommended.

4. A descriptive bulletin describing insulation offered.
5. Recommended lubricant specification.
6. Guaranteed noise limits.
7. List of special tools and gauges.
8. Time required after order placement for supply of certified outline and other drawings and pertinent engineering information.
9. Vendor shall state in writing compliance with all applicable specifications or list all exceptions thereto in a single section of his proposal entitled "EXCEPTIONS". Acceptance of a proposal shall not be considered a waiver of any specified requirements unless specific exception has been taken by the Vendor
10. The Vendor shall indicate in his quotation all materials offered are asbestos free.

ATTACHMENT 2
DATA REQUIRED AFTER PLACEMENT OF ORDER

1. All data and information covered by the Inquiry and the Material Requisition.
2. For motors:
   a) Time-current curves showing the allowable starting conditions and continuous operation at rated voltage and rated ambient temperature.
   b) Speed-torque and speed-current curves at 100% and 80% rated voltage.
   c) Saturation factor allowed for those motors whose starting current and torque characteristics have to be measured at reduced voltage.
   d) Data and drawings required by driven machine and/or gear
3. Certified copies of test data shall be submitted to IOCL prior to shipment.

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TOYO ENGINEERING INDIA PVT. LTD.	ATTACHMENT 4 HV INDUCTION MOTOR DATA SHEET	DOC NO. : IEGP-6351-8531-SP-000-1001
ISSUED : 9/02/2018	IEGP	PAGE 1 OF 1

Rev	MOTOR RATING		
	Rated output	KW	
	No. of poles / Speed	/	rpm
	Rated voltage & frequency	Refer Note 4 ± 6% V / 50 ± 3%	Hz
	Thermal classification	(X) F ( ) B ( )	
	Limit of temp. rise (Class)	( ) F (X) B ( )	
	Applicable codes & standards	(X) IEC 34 (X)IS 325	
	Type of motor		
	System Fault Level	- - KA for 1 Sec	

Rev	OPERATING CONDITIONS	
	Installation	(X) Outdoor ( ) Indoor
	Relative Humidity at T max	99.7%
	Design Ambient temperature	+45 °C
	Area classification	( ) Non-hazardous ( ) Hazardous
	(as per IS/IEC)	( ) Zone2, Temp. : T3 Gas Group : IIA ,IIB, IIC
	Tropicalisation	Required
	Corrosive atmosphere	( ) No (X) Yes

Rev	TYPE OF ROTOR	
	Type of rotor	(X) Squirrel cage ( ) Wound
	Method of cooling	(X) IC411 ( ) IC611 (X) TEFC
		( ) IC37 ( )
	Degree of protection	MOTOR : IP55 TB : IP55
	Type of protection	( ) "d" ( ) "p"
		( ) "e" ( ) "n"
	Mounting arrangement *	( ) IMB3 ( ) IMB5 ( ) IMV5
		( ) IMV1 ( )
	Shaft extension	(X) One extension
		( ) Two extension

Rev	OPERATION	
	Duty	(X) Continuous "S1" ( ) Intermittent "S "
	Starting method	
	Starting	(X) Loaded ( ) Unloaded
	starting current	Refer Note 3
	Conditions	( ) Reacceleration ( )
	Direction of rotation	( ) CW ( ) Bidirectional
	(Facing the shaft end)	( ) CCW
		( ) Reversible

Rev	TERMINAL BOX & TERMINALS	
	Position (Facing the shaft	( ) Right (X) Left
	end)	( ) Top
	Earth terminals	terminals diagonally opposite point on Motor + one inside terminal b

Rev	Gland Entry	Cable type/Size	Gland thread size		Cable dia (In/Out)
	for motor main			Metric	mm / mm
	for space heater			Metric	mm/ mm
	for temp. detector			Metric	mm/ mm

Rev	DRIVEN MACHINE DATA *		
	Manufacturer's name		
	Kind of machine		
	Absorbed power		BkW
	Required speed		rpm
	Moment of inertia(Note-3)	wk^2 =	kg.m^2
	Direction of Rotation		
	Type of Mounting		
	Thrust load	( ) Up	N, ( ) Down N
	Coupling arrangement	( ) Direct	( ) Gear
		( ) V-Belts	( )

Rev	NOTES
	1. starting time calculation shall be based on operating conditions such as open valve
	condition/closed valve condition, at no load /under load as applicable
	2. In case of gear box is provided , equipment wk^2 value at gear box input shaft shall be specified.
	3. For Starting Current Limitation, of the motors Please refer Electrical Design Data
	4. For voltage rating please refer Electrical Design Data

Rev	MANUFACTURER'S DATA *			
	Manufacturer's name			
	Type designation			
	Frame No.			
	Rated output / Voltage	kW /		V
	Rated speed			rpm
	Certifying authority,if motor is designed for classified area			
	Degree of protection	MOTOR : IP	TB : IP	
	Type of protection			
	Mounting arrangement (IM code)			
	Test according to			
	Load	1/2	3/4	FL
	Current (A)			
	Efficiency (%)			
	Power factor			
	Full load torque			N.m
	Break down torque			%
	Pull-up torque			%
	Locked rotor torque			%
	Locked rotor current (At 80%,100% rated volatge,reduced voltage starting)			%
	Locked rotor power factor(At 80% and 100% rated volatge, reduced voltage starting)			%
	Locked rotor time (t_E) (At 80%,100% rated volatge,at reduced voltage starting)			
	Start. time at 80% of rated voltage			sec.
	Start. time at 60% of rated voltage - for soft starters			sec.
	Rotor weight			sec.
	Winding connection			kg
	Total weight			kg
	Heating time constant and cold time constant			sec
	Moment of inertia wk^2			kg.m^2
	Bearing		( ) Ball or Roller	( ) Sleeve
	Bearing No.		DE :	NDE :
	Lubricant			
	Negetive sequence current and withstand time		% of Ifull	Sec.
	Forced oil press. at inlet flange			MPa
	Forced oil Q'ty per bearing			l/min
	Permissible shaft end float			mm
	Motor Terminal Box		Type	No of Terminals
	Aux Terminal Box		Type	No of Terminals
	Fault Level Withstand Time and Current for TB		KA	S
	Max. sound press. level (no load)			dB(A)
	Cooling time constant			
	Heating time constant			

Rev	ACCESSORIES
	(X) Space heater (240 V, * W)
	(X) Base
	(X) Shaft key
	(X) Stud terminals
	( ) Cable gland
	( ) Tinned Copper Lugs at Cable termination
	(X) Anchor bolts
	(X) Canopy for outdoor motors
	(X) Neutral Terminal Box
	(X) Motor Differential Protection CTs (Only for motors rated 1500kW and above)
	(X) RTDs - 6 Nos. Duplex type RTDs
	(X) BTDs - 2 Nos. Dial type thermometer shall be provided in case of sleeve bearing

Rev	ADDITIONAL INFORMATION BY VENDOR
	(X) Outline drawings
	(X) Motor speed - torque curve (At 80%,100% of rated voltage and reduced voltage
	(X) Motor speed - current curve (At 80%,100% of rated voltage and reduced voltage
	(X) Motor time - current heating curve(At 80%,100% of rated voltage and reduced voltage starting)
	(X) Motor speed - Time curve (At 80%,100% of rated voltage and reduced voltage
	(X) Performance Curve (P.F., Efficiency,Slip etc.)
	(X) Thermal withstand curve (At 80%,100% of rated voltage and reduced voltage startin
	(X) Negative sequence withstand characteristic
	(X) Starting time calculations

Rev	SPECIAL REQUIREMENT
	(X) Current Pulsation Analisys (for Reciprocating Type Driven Machine)
	( ) Torsional Analysis

(X) : INQUIRED or APPLIED * : TO BE SPECIFIED BY VENDOR

TOYO ENGINEERING INDIA PVT. LTD.	ATTACHMENT 3 LV INDUCTION MOTOR DATA SHEET	DOC NO. : IEGP-6351-8531-SP-000-1001
ISSUED : 9/02/2018	IEGP	PAGE 1 OF 1

Rev	MOTOR RATING	
	Rated output	KW
	No. of poles / Speed	/ rpm
	Rated voltage & frequency	415 ± 6% V / 50 ± 3% Hz
	Thermal classification	(X) F ( ) B ( )
	Limit of temp. rise (Class)	( ) F (X) B ( )
	Applicable codes & standards	(X) IEC 34 (X)IS 325
	Type of motor	
	System Fault Level	- - KA for 1 Sec

Rev	OPERATING CONDITIONS	
	Installation	(X) Outdoor ( ) Indoor
	Relative Humidity at T max	99.7%
	Design Ambient temperature	+45 °C
	Area classification	( ) Non-hazardous ( ) Hazardous
	(as per IS/IEC)	( ) Zone2, Temp. : T3 Gas Group : IIA ,IIB, IIC
	Tropicalisation	Required
	Corrosive atmosphere	( ) No (X) Yes

Rev	TYPE OF ROTOR	
	Type of rotor	(X) Squirrel cage ( ) Wound
	Method of cooling	(X) IC411 ( ) IC611 (X) TEFC
		( ) IC37 ( )
	Degree of protection	MOTOR : IP55 TB : IP55
	Type of protection	( ) "d" ( ) "p"
		( ) "e" Ref. Note ( ) "n"
	Mounting arrangement *	( ) IMB3 ( ) IMB5 ( ) IMV5
		( ) IMV1 ( )
	Shaft extension	(X) One extension
		( ) Two extension

Rev	OPERATION	
	Duty	(X) Continuous "S1" ( ) Intermittent "S "
	Starting method	(X) Direct-on-line
	Starting	(X) Loaded ( ) Unloaded
	starting current	Refer Note 3
	Conditions	( ) Reacceleration ( )
	Direction of rotation	( ) CW ( ) Bidirectional
	(Facing the shaft end)	( ) CCW
		( ) Reversible

Rev	TERMINAL BOX & TERMINALS	
	Position (Facing the shaft end)	( ) Right (X) Left
		( ) Top
	Earth terminals	2 terminals diagonally opposite point on Motor

Rev	Gland Entry	Cable type/Size	Gland thread size		Cable dia (In/Out)
	for motor main			Metric	mm / mm
	for space heater			Metric	mm/ mm

Rev	DRIVEN MACHINE DATA *	
	Manufacturer's name	
	Kind of machine	
	Absorbed power	BkW
	Required speed	rpm
	Moment of inertia(Note-3)	wk^2 = kg.m^2
	Direction of Rotation	
	Type of Mounting	
	Thrust load	( ) Up N, ( ) Down N
	Coupling arrangement	( ) Direct ( ) Gear
		( ) V-Belts ( )

Rev	NOTES
	1. starting time calculation shall be based on operating conditions such as. open valve condition/closed valve condition, at no load /under load as applicable
	2. In case of gear box is provided , equipment wk^2 value at gear box input shall be specified.
	3. For starrting Current Requirement please refer Electrical Design Data

Rev	MANUFACTURER'S DATA *			
	Manufacturer's name			
	Type designation			
	Frame No.			
	Rated output / Voltage	kW / V		
	Rated speed	rpm		
	Certifying authority,if motor is designed for classified area			
	Degree of protection	MOTOR : IP	TB : IP	
	Type of protection			
	Mounting arrangement (IM code)			
	Test according to			
	Load	1/2	3/4	FL
	Current (A)			
	Efficiency (%)			
	Power factor			
	Full load torque	N.m		
	Break down torque	%		
	Pull-up torque	%		
	Locked rotor torque	%		
	Locked rotor current (At 80% and 100% rated volatge)	%		
	Locked rotor power factor(At 80% and 100% rated volatge)	%		
	Locked rotor time (t_E) (At 80% and 100% rated volatge)			
	Start. time at 80% of rated voltage	sec.		
	Rotor weight	sec.		
	Winding connection	kg		
	Total weight	kg		
	Heating time constant and cold time constant	sec		
	Moment of inertia wk^2	kg.m^2		
	Bearing	( ) Ball or Roller		
	Bearing No.	DE : NDE :		
	Lubricant			
	Negetive sequence current and withstand time	% of Ifull Sec.		
		MPa		
		l/min		
	Permissible shaft end float	mm		
	Motor Terminal Box	Type No of Terminals		
	Aux Terminal Box	Type No of Terminals		
	Fault Level Withstand Time and Current for TB	KA S		
	Max. sound press. level (no load)	dB(A)		

Rev	ACCESSORIES
	(X) Space heater (240 V, * W)
	(X) Base
	(X) Shaft key
	(X) Stud terminals
	( ) Cable gland
	( ) Tinned Copper Lugs at Cable termination
	(X) Anchor bolts
	(X) Canopy for outdoor motors
	( ) Neutral Terminal Box

Rev	ADDITIONAL INFORMATION BY VENDOR
	(X) Outline drawings
	(X) Motor speed - torque curve (At 80% & 100% of rated voltage)
	(X) Motor speed - current curve (At 80 % and 100% of rated voltage)
	(X) Motor time - current heating curve (At, 80 % and 100% of rated voltage)
	(X) Motor speed - Time curve (At80 % and 100% of rated voltage)
	(X) Performance Curve (P.F., Efficiency,Slip etc.)
	(X) Thermal withstand curve (At 80 % and 100% of rated voltage)
	(X) Negative sequence withstand characteristic
	(X) Starting time calculations

Rev	SPECIAL REQUIREMENT
	(X) Current Pulsation Analysis (for Reciprocating Type Driven Machine)
	( ) Torsional analysys

(X) : INQUIRED or APPLIED * : TO BE SPECIFIED BY VENDOR